

Cross-Material Transfer of GLAD Column-Tilt and Porosity Models: A Rigorous Negative Result and Its Statistical Boundary

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Abstract

Glancing-angle deposition (GLAD) produces tilted, porous columnar thin films whose column-tilt angle β and porosity ϕ are the two morphological quantities most directly linked to a film’s optical and electrical anisotropy. Existing predictive relations for both quantities are either purely geometric (the tangent rule for β) or material-specific empirical fits; no published model uses material identity itself – through physically motivated descriptors such as atomic mass, melting point, or bulk density – to predict how a *new* material, never seen during fitting, will behave under GLAD. We assembled a literature-derived corpus of experimentally measured $\beta(\alpha)$ and porosity(α) values spanning up to 40 distinct materials (elemental metals, oxides, fluorides, chalcogenides), fitted four candidate models (linear regression, Gaussian process regression, gradient-boosted trees, random forest) under leave-one-material-out cross-validation (LOMO-CV), and pre-registered a strict acceptance bar before inspecting any result: $\geq 15\%$ LOMO-CV RMSE improvement over a physics/naive baseline *and* a material-level permutation-test $p < 0.05$. Every learned model comfortably clears the RMSE bar for $\beta(\alpha)$ (up to +54.4% depending on corpus snapshot) and often for porosity(α) (up to +28.5%), but no configuration – the pooled fit, three pre-registered material-class subgroups, two independent corpus-growth interventions targeting the most promising subgroup, a pairwise ranking reformulation of the task, or a pre-registered Fisher’s-method combination across the three disjoint material-class subgroups ($p = 0.715$) – ever clears the permutation-test bar (best $p = 0.09$ for $\beta(\alpha)$ at an earlier, smaller corpus snapshot; $p = 0.10$ – 0.86 across every variant tested at the current snapshot, with the pooled $\beta(\alpha)$ and porosity(α) fits – $p = 0.10$ and $p = 0.12$ – now the closest either has come to significance across roughly eleven corpus-growth rounds). We report this as an honest, thoroughly instrumented negative result: the barrier is specifically material-level sample size, not model choice, feature choice, task formulation, or extraction quality, all four of which were tested and ruled out as explanations. As a complementary, deliberately different analysis, a pre-registered Bayesian hierarchical partial-pooling model fitted to the same $\beta(\alpha)$ corpus ($n = 213$ rows, 40 materials, all MCMC diagnostics clean) finds that atomic mass has a credible non-

zero effect on a material’s mean offset from the tangent rule (94% highest density interval excluding zero) and that Bayesian R^2 posteriors place 80–92% probability on material descriptors explaining more than 10% of between-material variance. This does not overturn the frequentist non-significance verdict – it explains it: a real-but-modest effect of the size the Bayesian model finds is exactly what a permutation test at $n = 14$ –43 materials is underpowered to detect at $p < 0.05$. We present this paper as a resource for the GLAD/materials-informatics community: a fully pre-registered, diagnostically exhaustive account of where the statistical-power boundary for this class of cross-material generalization problem currently sits, together with the specific corpus-growth target (professional literature-database access to 30+ elemental metals, particularly platinum-group and rare-earth species) that would most directly move it.

1 Introduction

Glancing-angle deposition (GLAD) is a physical vapor deposition technique in which the substrate is tilted at a large angle α (measured from the surface normal) relative to the incoming vapor flux, so that geometric self-shadowing among growing nuclei produces isolated, tilted columnar structures rather than a dense film [Robbie and Brett, 1997, Hawkeye and Brett, 2007, Hawkeye et al., 2014]. The resulting films combine two properties of independent technological interest: a column-tilt angle β , generally smaller than the deposition angle α because incoming flux is captured preferentially by the upper, unshadowed part of a growing column, and an internal porosity ϕ set by how completely that self-shadowing excludes flux from the space between columns. Both properties drive the strong optical and electrical anisotropy that makes GLAD films attractive for polarization optics, sensing, and photovoltaic light management [Brett and Hawkeye, 2008, Lakhtakia and Messier, 2005, Messier et al., 2000].

The relationship between α and β has a long-standing geometric approximation, the tangent rule [Tait et al., 1993, Thornton, 1974], which treats column growth as a purely ballistic shadowing process independent of the depositing material’s identity. In practice, measured $\beta(\alpha)$ curves for different materials deposited under nominally comparable conditions show real, reproducible scatter around the tangent-rule prediction [Chaffar Akkari et al., 2018, 2019, Tounsi et al., 2015, Benamor et al., 2026, Ben Nacer et al., 2020] – a scatter plausibly attributable to material-dependent surface mobility, sticking coefficient, and crystal structure, none of which the tangent rule encodes. Porosity has no comparably established closed-form geometric rule at all; published porosity(α) values are almost always reported and modeled per-material. This project’s own ballistic GLAD simulator (companion manuscript, Ben Haj Ali et al., 2026a) and density-calibrated neural-field surrogate (companion manuscript, Ben Haj Ali et al., 2026b) both operate at fixed, explicitly specified material parameters; a model that could predict $\beta(\alpha)$ or $\phi(\alpha)$ for a material *not* already in the simulator’s parameter database, using only readily available descriptors of that material, would be directly useful for screening new material choices before running any simulation or deposition campaign.

This motivates a natural question: can material descriptors (atomic mass, melting point, bulk density, crystal structure, deposition method) predict a material’s $\beta(\alpha)$ or $\phi(\alpha)$ curve well enough, and reliably enough across held-out materials, to be useful for this kind of screening? We built a literature-derived corpus spanning up to 40 distinct materials across elemental metals, oxides, fluorides, and chalcogenides, and answered this question with a pre-registered, leakage-controlled leave-one-material-out (LOMO-CV) evaluation protocol, reported here in full together with the extensive, honestly documented attempts made to recover a positive result once the pooled result fell short. We regard the resulting negative-to-mixed finding as a genuine contribution in its own right, not a shortfall to be minimized: it identifies exactly where, quantitatively, the statistical-power boundary for this class of materials-informatics problem currently sits, rules out four plausible alternative explanations (wrong features, wrong task formulation, class heterogeneity, extraction quality) for the boundary, and provides a Bayesian complementary analysis that characterizes the size of the effect the frequentist test cannot yet resolve. This is offered explicitly as a resource for future corpus-growth efforts in this specific sub-field, in the spirit that a well-instrumented negative result that saves the next researcher from repeating the same five failed levers is worth reporting on its own merits.

The remainder of this paper is organized as follows. Section 2 describes the corpus, the pre-registered evaluation protocol, the four candidate models, the material-class stratification, the pairwise-ranking reformulation, and the Bayesian hierarchical model specification. Section 3 reports the pooled results for all four target observables, the class-stratified results, the targeted corpus-growth push within the most promising class, the ranking reformulation, and the Bayesian hierarchical analysis. Section 4 interprets these results together, argues for reading them as a genuine statistical-power finding rather than a failure of modeling approach, and identifies the concrete corpus-growth target that would most directly resolve the open question. Section 5 summarizes.

2 Methods

2.1 Corpus and target observables

Four target observables were evaluated, all as a function of deposition angle α : column-tilt angle $\beta(\alpha)$, porosity $\phi(\alpha)$, refractive index $n(\alpha)$, and optical band gap $E_g(\alpha)$. Rows were drawn from the project’s ongoing literature-absorption effort (peer-reviewed papers and, in a small number of cases, PhD theses, reporting angle-resolved measurements of one or more of these quantities for a named material) and assembled into per-observable feature tables by an automated pipeline (`build_feature_table.py`) that attaches five candidate material descriptors to every row: atomic mass, melting point, bulk density, crystal structure (categorical), and deposition method (categorical: e-beam evaporation, thermal evaporation, DC/RF sputtering, etc.). Rows lacking a numeric target value, an unresolved deposition angle, or a required descriptor for the base material were excluded (26/228 excluded rows for β , 11/114 for porosity, at the snapshot reported in Table 1); exclusion reasons are logged per-row and are not adjusted after seeing model results. Corpus size grew substantially over the course of this investigation as literature absorption continued in parallel with model evaluation; every result below states the exact $(n_{\text{rows}}, n_{\text{materials}})$ snapshot it was computed on, since corpus snapshots differ across sections of this paper by design – later sections report tests run against a larger corpus than earlier sections, reflecting real, ongoing corpus growth rather than selective reporting. The most recent growth (2026-08-20) added a full packing-density series for Nb_2O_5 [Xiao et al., 2008] and a full column-tilt-angle series for Ta_2O_5 [Sobahan et al., 2009] – both materials previously missing that specific observable entirely – ten further β rows for TiN from a second literature source [Bouaouina et al., 2018] (TiN itself was already in the corpus from an earlier source, so this is deeper coverage of an existing material, not a new one), and one new elemental metal, Pd [Kim et al., 2005] ($\beta = 62^\circ$ at $\alpha = 85^\circ$, a single data point flagged and treated the same way as the existing single-point Sb entry, not yet incorporated into the elemental-metals class-stratified fit of Section 3.3). A second growth round, folded in immediately afterward (2026-08-20/21) from a larger literature batch, added two further

new materials – CeO_2 [Hodgkinson et al., 1996], with a six-point column-tilt series ($0\text{--}45^\circ$) that is non-monotonic and disagrees markedly with the tangent-rule prediction over the same range, and LaF_3 [Liu et al., 2007] – plus new-source or new-observable rows for existing materials: a TiO_2 band-gap series [Pandey et al., 2022] (TiO_2 ’s first E_g entry), CdTe column-tilt and band-gap rows from a second source [Ehsani et al., 2013], ZnS and MgF_2 endpoint rows for porosity and refractive index [Oh and Kim, 2020], a third ZnS column-tilt/band-gap source [Molla Ali Ashrafi et al., 2018], and SnS refractive index and band-gap rows [Sazideh et al., 2017, 2018] (SnS previously had only column-tilt and porosity coverage). All pooled and per-observable results in Section 3 are computed on the corpus snapshot after both growth rounds.

Table 1: Feature-table coverage at the most complete raw-table snapshot produced during this investigation (2026-08-17), shown here to illustrate typical exclusion reasons and per-observable coverage. Because literature absorption continued throughout the investigation, this raw snapshot is not identical to any single downstream analysis’s corpus: the pooled baseline (Section 3.1), the earlier corpus-growth campaign (Table 4), the class-stratification and ranking analyses (Sections 3.3–3.6), and the Bayesian hierarchical model (Section 3.7) were each run at their own specific, smaller snapshot along this same growth trajectory (after an analysis-specific descriptor-completeness filter), and each result below states its own exact $(n_{\text{rows}}, n_{\text{materials}})$.

Target observable	Rows	Distinct materials
Column-tilt angle $\beta(\alpha)$	228 (202 usable)	43
Porosity $\phi(\alpha)$	114 (103 usable)	23
Refractive index $n(\alpha)$	105 (102 usable)	16
Band gap $E_g(\alpha)$	78 (72 usable)	12

2.2 Pre-registered evaluation criterion

Before fitting any model to a given corpus snapshot, a single acceptance criterion was fixed for each target observable, independent of the outcome: a candidate model was considered to have cleared the bar only if it achieved (i) a leave-one-material-out cross-validated RMSE at least 15% lower than the corresponding baseline (the tangent rule for $\beta(\alpha)$; the LOMO-CV train-mean for porosity, refractive index, and E_g , since no established closed-form geometric rule exists for these), and (ii) a material-level permutation-test p -value below 0.05. Both conditions were fixed prior to inspecting either result and applied

identically across all four observables and every corpus snapshot reported below. Results are reported as obtained, including every negative and partially negative outcome, without post-hoc adjustment of the criterion.

2.3 Leave-one-material-out cross-validation

All model comparisons use leave-one-material-out (LOMO-CV) rather than leave-one-row-out cross-validation: rows from the same material (often multiple angles from a single reproducibility series or multiple points digitized from one figure) are not statistically independent, and evaluating a model on a held-out *row* whose material was seen during training would leak information the model could exploit without any genuine cross-material generalization ability. Under LOMO-CV, one material’s entire row set is held out as the test fold on each iteration, and the model is refit on the remaining materials; this directly measures whether descriptor-based generalization to a truly unseen material is possible, which is the scientific question this paper asks.

2.4 Candidate models

Four models were compared under the identical LOMO-CV protocol, plus the relevant baseline for each observable:

- **Physics/naive baseline:** the tangent rule (a zero-free-parameter geometric relation) for $\beta(\alpha)$; the LOMO-CV train-mean for porosity, refractive index, and E_g (no established closed-form rule for these).
- **Linear regression** on the standardized descriptor set.
- **Gaussian process regression** (Matérn-5/2 kernel plus a white-noise term; `scikit-learn` `GaussianProcessRegressor`) [Rasmussen and Williams, 2006, Pedregosa et al., 2011] – designated the primary candidate at the project’s earliest evaluation round, before the gradient-boosted and random-forest comparators were added.
- **Gradient-boosted trees** [Friedman, 2001, Pedregosa et al., 2011].

- **Random forest** [Breiman, 2001, Pedregosa et al., 2011].

All descriptor-based models share the same feature pipeline: numeric descriptors are standardized (zero mean, unit variance, fit on the training fold only) and categorical descriptors (crystal structure, deposition method) are one-hot encoded, both wrapped in a `scikit-learn` `ColumnTransformer`/`Pipeline` so that no fold-specific statistic is computed on held-out data.

2.5 Material-level permutation test

Statistical significance of a model’s LOMO-CV RMSE improvement over its baseline is assessed with a material-level permutation test [Ojala and Garriga, 2010]: the material-block labels attached to each row (not individual rows) are shuffled $n_{\text{perm}} = 100$ times, the same LOMO-CV model-fitting and RMSE-improvement calculation is repeated on each shuffled assignment, and p is the fraction of shuffles whose RMSE improvement equals or exceeds the improvement observed on the real, unshuffled data. Shuffling at the material-block level, rather than the row level, is essential: row-level shuffling would break the same within-material correlation structure that motivates LOMO-CV in the first place and would understate the true null distribution’s spread, inflating apparent significance. An early corpus snapshot ($n = 73$ rows, 23 materials) returned $p = 0.020$ for $\beta(\alpha)$, initially reported as a cleared result; this did not survive further, real corpus growth (Section 3.1), which is itself part of this paper’s evidence that a single small-corpus permutation-test result should not be treated as final without a subsequent replication check as the corpus grows.

2.6 Material-class stratification

To test whether the pooled model’s failure to clear significance reflects genuine cross-class physical heterogeneity being averaged away – elemental metals, ionic/covalent oxides and fluorides, and covalent chalcogenides plausibly respond differently to ballistic shadowing – a grouping by crystal-structure/bonding chemistry was pre-registered and frozen *before* any per-class fit was run: elemental metals (fcc/bcc/hcp pure elements), rock-salt nitrides

(grouped with AlN by anion chemistry), oxides/fluorides (ionic/covalent, ZnO included by anion chemistry despite its wurtzite structure), and chalcogenides/sulfides. A minimum of five materials per subgroup was pre-registered as a floor for testing; subgroups below this floor are reported as untested, not as null results. Each qualifying subgroup was fit with the identical four-model comparison, LOMO-CV-CV, and material-level permutation protocol used for the pooled model.

2.7 Pairwise ranking reformulation

As an orthogonal test of whether the pre-registered exact-value regression criterion is testing the wrong task, a pairwise-ranking reformulation was pre-registered (RANKING_REFORMULATION_PREREGISTRATION_20260817.md, fixed before fitting) and fit to the $\beta(\alpha)$ corpus using PyMC 5.28.5 with ArviZ 0.23.4 for diagnostics [Salvatier et al., 2016], sampled with the No-U-Turn Sampler (NUTS, CPU only). This model asks a different question from the frequentist tests above: not “does a flexible model out-predict the physics baseline on a held-out material by a margin unlikely under label-shuffling?”, but “how much of the material-to-material spread in $\beta(\alpha)$ is attributable, in a

2.8 Bayesian hierarchical partial-pooling model

As a final, deliberately different analysis, a Bayesian hierarchical (partial-pooling) model was pre-registered (BAYESIAN_HIERARCHICAL_PREREGISTRATION_20260817.md, fixed before fitting) and fit to the $\beta(\alpha)$ corpus using PyMC 5.28.5 with ArviZ 0.23.4 for diagnostics [Salvatier et al., 2016], sampled with the No-U-Turn Sampler (NUTS, CPU only). This model asks a different question from the frequentist tests above: not “does a flexible model out-predict the physics baseline on a held-out material by a margin unlikely under label-shuffling?”, but “how much of the material-to-material spread in $\beta(\alpha)$ is attributable, in a

regularized partial-pooling regression, to physically motivated descriptors, and with what uncertainty?” The model places a per-material random intercept a_i and slope b_i on the tangent-rule residual (i.e., a material-specific affine correction to the physics baseline as a function of α), regressed at the population level on three standardized descriptors (atomic mass, melting point, bulk density); crystal structure was excluded from the group-level regression for identifiability reasons at 19 levels across 43 materials. Convergence was assessed with the rank-normalized $\hat{R}$ diagnostic [Vehtari et al., 2021] and effective sample size. The primary deliverable is the Bayesian R^2 [Gelman et al., 2019] – computed per posterior draw as $\text{Var}(\text{fitted})/(\text{Var}(\text{fitted}) + \text{residual variance})$, separately for the intercept layer (R_a^2) and the slope layer (R_b^2) – reported as an in-sample quantity (how much structure the fitted model captures), not a cross-validated one (how well it would predict a new material).

3 Results

3.1 Pooled cross-material models: RMSE bar cleared, significance bar not

Table 2 summarizes the best-model LOMO-CV result for all four target observables at the latest corpus snapshot ($\beta(\alpha)$: $n = 250$ rows/46 materials; porosity(α): $n = 108/24$; refractive index: $n = 78/13$; E_g : $n = 67/10$). For $\beta(\alpha)$, the tangent-rule baseline achieves a LOMO-CV RMSE of 25.522° ; every learned model outperforms it, with gradient-boosted trees the best performer at 12.773° (+50.0%). This comfortably clears the pre-registered 15% RMSE bar. The material-level permutation test, however, returns $p = 0.10$ ($n_{\text{perm}} = 100$), still short of the 0.05 significance threshold – but the closest the pooled $\beta(\alpha)$ fit has come to significance across roughly eleven independent corpus-growth rounds (was $p = 0.20$, then $p = 0.22$ one round earlier). For porosity(α), Gaussian process reaches +20.7% RMSE improvement over the naive-mean baseline (24.315 vs. 30.667, porosity in percentage points) – clearing the RMSE bar – with permutation $p = 0.12$, likewise the closest this fit has come to significance (was $p = 0.32$, then $p = 0.36$). One of the

284 five porosity candidates, linear regression, underperforms the baseline outright (-24.9%);
 285 the remaining three learned models cluster closely together and all clear the RMSE bar
 286 (Gaussian process $+20.7\%$, the best performer; random forest $+19.5\%$; gradient-boosted
 287 trees $+18.8\%$), underscoring that the RMSE improvement Gaussian process achieves is a
 288 genuine, model-class-independent effect and not an artifact of a single weak comparator.
 289 Refractive index ($+16.2\%$, $p = 0.54$) shows the same pattern. Band gap $E_g(\alpha)$ is the sole
 290 observable for which no learned model even clears the RMSE bar (best learned model
 291 -9.3% , i.e. worse than baseline; the naive-mean baseline itself remains the best “model”),
 292 so its permutation test is not applicable.

293 Both p -value drops (beta: $0.22 \rightarrow 0.20 \rightarrow 0.10$; porosity: $0.36 \rightarrow 0.32 \rightarrow 0.12$) tracked
 294 a genuine, incremental corpus-growth process rather than any change in method or re-
 295 analysis of existing data – each round added new literature-sourced rows and, in this
 296 latest round, two genuinely new materials to the corpus (cerium oxide and lanthanum
 297 fluoride, Section 3.4). Roughly halving both p -values with a handful of added materials is
 298 consistent with, and strengthens, this paper’s central diagnosis that material-level sample
 299 size, not model choice or task formulation, is the binding constraint: the trend moves in
 300 exactly the direction that diagnosis predicts. We report it plainly as what it is – a trend,
 301 not a reversal – since neither pooled fit clears the pre-registered $p < 0.05$ bar.

Table 2: Best-model LOMO-CV-CV comparison against the pre-registered bar, all four
 target observables, latest corpus snapshot. RMSE improvement is relative to the tangent-
 rule baseline ($\beta(\alpha)$) or the LOMO-CV train-mean baseline (porosity, n , E_g). Permutation
 p from $n_{\text{perm}} = 100$ material-block shuffles.

Observable	Rows	Materials	Best model	RMSE Δ	Perm. p	Bar cleared?
$\beta(\alpha)$	250	46	gradient-boosted trees	$+50.0\%$	0.10	No
Porosity $\phi(\alpha)$	108	24	Gaussian process	$+20.7\%$	0.12	No
$n(\alpha)$	78	13	Gaussian process	$+16.2\%$	0.54	No
$E_g(\alpha)$	67	10	naive-mean baseline	$+0.0\%$	n/a	No

302 3.1.1 Feature importance for the confirmed $\beta(\alpha)$ model

303 A leave-one-feature-out (LOFO) analysis on the primary Gaussian process candidate at an
 304 earlier corpus snapshot ($n = 73$ rows/23 materials, full-model LOMO-CV RMSE = 11.593°)

shows the deposition angle α itself dominates, as expected (RMSE rises to 19.363° when removed). Among material descriptors, melting point (RMSE 13.980° without it) and atomic mass (12.585°) carry essentially all of the genuine cross-material signal beyond α ; bulk density and crystal structure do not merely fail to help – removing either one *improves* LOMO-CV RMSE slightly (to 11.460° and 10.793° respectively), indicating these two features contribute noise rather than signal to the fitted GP at this corpus size (Table 3). This descriptive analysis is not itself subject to the pre-registered significance bar (which applies only to the model-vs-baseline comparison), but it is directly consistent with the Bayesian hierarchical model’s independent identification of atomic mass as the one descriptor with a credibly non-zero effect (Section 3.7).

Table 3: Leave-one-feature-out importance for the $\beta(\alpha)$ Gaussian process model ($n = 73$ rows, 23 materials). A larger RMSE increase when a feature is removed indicates more predictive signal currently attributed to that feature.

Feature removed	LOMO-CV RMSE without it ($^\circ$)	Δ RMSE
α (deposition angle)	19.363	+7.769
Melting point	13.980	+2.387
Atomic mass	12.585	+0.991
Deposition method	11.819	+0.225
(<i>full model</i>)	11.593	—
Bulk density	11.460	−0.134
Crystal structure	10.793	−0.801

3.2 The corpus-growth campaign: RMSE improves, significance does not

Following the initial cleared result at $n = 73$ rows/23 materials ($p = 0.020$), further, real corpus growth caused the permutation p -value to rise, first to 0.13 at $n = 93/25$ materials, confirmed at a second random seed ($p = 0.09$). Five independent corpus-growth avenues were then pursued deliberately, rather than treating this as a one-off setback: a targeted literature-backlog absorption pass; a web-escalation search for entirely new source materials (finding two, SiO_2 and HfO_2 , with genuine open-access angle-series data, and correctly excluding two paywalled/snippet-only leads, Ta_2O_5 and TiN , rather than fab-

ricating values from partial information); a feature-engineering ablation adding Cordero covalent atomic radius and homologous temperature T/T_{melt} ; a deep re-digitization of nine already-absorbed source papers under a stricter protocol of visually analyzing every figure directly rather than relying on text extraction (which recovered 43 new data points, most notably growing ZnO’s series from 1 to 22 points); and a whole-drive sweep for misplaced literature PDFs (which found one further relevant paper, reporting only two discrete angles, too sparse to contribute LOMO-CV rows). Table 4 reports the resulting best figures at $n = 132$ rows/27 materials.

Table 4: Corpus-growth campaign result, all four observables, at the $n = 132$ rows/27-materials $\beta(\alpha)$ snapshot (the single most effective lever tried was the deep figure-level re-digitization pass).

Observable	Rows	Materials	Best model	RMSE Δ	Perm. p	Bar cleared?
$\beta(\alpha)$	132	27	gradient-boosted trees	+46.4%	0.09	No
Porosity $\phi(\alpha)$	64	13	random forest	+28.5%	0.40	No
$n(\alpha)$	37	7	linear regression	+13.1%	0.70	No (fails RMSE too)
$E_g(\alpha)$	49	6	naive-mean baseline	—	n/a	No

The feature-engineering ablation moved neither RMSE improvement nor permutation p for either β or porosity, ruling out “wrong feature set” as an explanation for the shortfall. $\beta(\alpha)$ ’s permutation p fell monotonically with every genuine data-density increase across this round of the investigation ($0.13 \rightarrow 0.09$), a directional trend more consistent with a real, currently under-sampled effect than with pure noise – offered here as a plausibility argument, not as evidence the effect is proven; the honest reportable result at this snapshot remained NOT CLEARED.

3.3 Does material-class heterogeneity explain the pooled failure?

By a later corpus snapshot – $n = 37$ materials for the pooled $\beta(\alpha)$ fit, gradient-boosted trees +49.2%, $p = 0.22$ (the snapshot used for this stratification analysis; the pooled fit in Table 2 reflects a subsequently grown corpus, $n = 46$ materials, $p = 0.10$) – a natural hypothesis was tested directly: perhaps pooling elemental metals, oxides/ fluorides, and

chalcogenides into a single fit averages away real, class-specific physics that a per-class fit would recover. Table 5 reports the pre-registered per-class result. The hypothesis is **not confirmed**: no class clears $p < 0.05$. Chalcogenides/sulfides reach +23.1% RMSE improvement at $p = 0.86$; oxides/fluorides +41.8% at $p = 0.55$; elemental metals – the largest and chemically most homogeneous subgroup – +44.6% at $p = 0.20$, the closest of the three to significance but still far above the bar. Rock-salt nitrides (4 materials, 18 rows) fell below the pre-registered five-material minimum and were not tested.

Table 5: Material-class-stratified $\beta(\alpha)$ results, pre-registered grouping by crystal-structure/bonding chemistry, identical four-model/LOMO-CV/permutation protocol as the pooled fit. Pooled reference row included for comparison.

Class	Materials	Rows	Best model	RMSE Δ	Perm. p	Bar
Rock-salt nitrides	4	18	—	—	—	Not tested
Chalcogenides/sulfides	6	38	random forest	+23.1%	0.86	
Oxides/fluorides	11	67	random forest	+41.8%	0.55	
Elemental metals	14	53	random forest	+44.6%	0.20	
<i>Pooled (all classes)</i>	46	250 ¹	gradient-boosted trees	+50.0%	0.10	

RMSE improvement is large in every class (23–45%), mirroring the pooled result: the models find *some* systematic structure in every partition tried, but that structure does not survive random re-partitioning of materials into cross-validation folds, in any grouping. This is not evidence against class-dependent physics – it means the current corpus cannot distinguish that hypothesis from ordinary small-sample noise, since every per-class subgroup (14, 11, or 6 materials) has fewer materials than the already-underpowered pooled set (37), trading away exactly the degrees of freedom the permutation test needs without a compensating gain in signal purity. Elemental metals’ comparatively low $p = 0.20$ was flagged as the most promising lead and pursued further.

¹Row count for the pooled reference row reflects the current Table 2 snapshot ($n = 46$ materials, 250 rows); the class-stratification analysis in this section was run at an earlier, smaller snapshot ($n = 37$ materials, 178 rows for the equivalent pooled comparison) and was not rerun against the later corpus growth. Per-class row counts above are the values actually used for the stratified fits.

3.4 A targeted corpus-growth push within elemental metals: also null

Two further, targeted literature-absorption interventions added real, sourced data specifically to the elemental-metals subgroup, run separately to isolate which kind of densification helps. **Cross-source densification** (adding 1–2 points from different labs/papers to each of the sparsest metals: W $1 \rightarrow 8$ points across two independent sources, Mo $1 \rightarrow 8$, Au $1 \rightarrow 4$) grew the subgroup to 72 rows at the same 14 materials. RMSE improvement continued to rise, to +52.7%, but p regressed, from 0.20 to 0.30. **Same-source densification** (digitizing Figure 11 of Besnard et al. [2025], a single consistent experimental campaign reporting nine transition metals on one angle series, adding three genuinely new materials – V, Hf, Ta – to the pre-registered elemental-metals class under the same fcc/bcc/hcp criterion decided before seeing any result) grew the subgroup to 17 materials, 88 rows, and gave $p = 0.24$ – better than the cross-source attempt but still worse than the untouched 14-material baseline, and far from 0.05. A subsequent, independent literature absorption (task #72, 2026-08-20) added Fe (bcc), the same fcc/bcc/hcp criterion applied before seeing this result, growing the subgroup to 18 materials, 90 rows, and giving $p = 0.33$ – again far from 0.05 and within the same 0.20–0.36 range observed across every other variant of this subgroup. Table 6 summarizes.

Table 6: Targeted corpus-growth interventions within the elemental-metals subgroup. Neither intervention moves p reliably toward significance; the range 0.20–0.36 across every variant is consistent with ordinary sampling noise around a true p that is not small at this sample size.

Variant	Materials	Rows	RMSE Δ (best model)	Perm. p
Baseline (Section 3.3)	14	53	+44.6%	0.20
Cross-source densification (W/Mo/Au)	14	72	+52.7%	0.30
Same-source densification (+V, Hf, Ta)	17	88	+54.4%	0.24
+ Fe (task #72 absorption, 2026-08-20)	18	90	+49.6%	0.33

Across every variant tried, p ranged 0.20–0.36, with no lever – more points on existing materials, new materials from a scattered set of sources, or new materials from one internally consistent source – moving it reliably toward significance. Recovering significance in this subgroup would need roughly double the current material count (30+ elemental

metals); a subsequent targeted search for 19 further elemental metals, mostly platinum-group and rare-earth species, returned usable new data for only one (Sb) via open web search – a genuine access limitation of the search tools available for this project, not proof the underlying literature does not exist.

3.5 A pre-registered meta-analytic combination

A final, orthogonal hypothesis is that the null result reflects per-test underpowering rather than a genuine absence of any effect – each individual test (pooled fit, each material-class subgroup, each observable) may be too small to detect a real but modest signal on its own, even if that signal is genuinely present and consistent in direction across tests. Fisher’s combined probability test ($X^2 = -2 \sum \ln p_i$, χ^2_{2k} under the joint null) is a standard tool for exactly this regime, and was applied here to two pre-registered combinations chosen for their independence properties, frozen before either was computed (full pre-registration: `META_ANALYSIS_PREREGISTRATION_20260820.md`).

The primary combination pools the three disjoint, pre-registered material-class subgroups from Section 3.3 – elemental metals ($p = 0.33$, updated after the Fe absorption of Section 3.4), oxides/fluorides ($p = 0.55$), and chalcogenides/sulfides ($p = 0.86$) – which partition the same corpus into non-overlapping material sets, the cleanest independence case available in this dataset. The combined statistic is $X^2 = 3.71$, $df = 6$, $p = 0.715$: **not cleared**. A secondary, explicitly exploratory combination pools the three pooled observable-level tests that individually cleared the RMSE bar – $\beta(\alpha)$ ($p = 0.10$, updated 2026-08-20/21 after the second same-day corpus-growth round, Section 3.1), porosity(α) ($p = 0.12$, same update), and $n(\alpha)$ ($p = 0.54$, unchanged) – for which the independence assumption is only approximate (materials and source papers overlap across observable corpora); this gives $X^2 = 10.08$, $df = 6$, $p = 0.121$ (recomputed by hand as $X^2 = -2 \sum \ln p_i = -2(\ln 0.10 + \ln 0.12 + \ln 0.54) = 10.078$, cross-checked against `scipy.stats.combine_pvalues(method='fisher')`; was $X^2 = 6.73$, $p = 0.347$ one round earlier), also not cleared but markedly closer than either prior snapshot of this combination.

Fisher’s method is most sensitive when several individual p-values cluster in a moderately small range (roughly 0.03–0.15) without any one crossing 0.05 alone. The current input set is closer to that regime than any earlier snapshot – two of the three inputs ($\beta(\alpha) = 0.10$, porosity = 0.12) now sit inside the sensitive band, with only $n(\alpha) = 0.54$ still far outside it – which is consistent with the combined p dropping by nearly a factor of three (0.347→0.121) even though no single test crossed 0.05 on its own. This still falls short of the pre-registered $p < 0.05$ bar, and remains fully consistent with the Bayesian hierarchical finding (Section 3.7), which concerns a different, narrower question (a specific credible effect of atomic mass within the elemental-metals corpus) rather than an aggregate cross-class or cross-observable claim.

3.6 Reformulating the task: ranking instead of regression

A separate, orthogonal hypothesis is that the pre-registered exact-value criterion tests the wrong task: predicting a material’s relative rank against another material at a comparable angle is a scientifically useful and potentially easier target than predicting its exact value. Table 7 reports the pre-registered pairwise-ranking result. For $\beta(\alpha)$ ($n = 213$ rows, 40 materials, 10,394 directed pairs within a $\pm 5^\circ$ angle window), the best model (a gradient-boosted ranker using the full descriptor set) reaches a pooled LOMO-CV pairwise accuracy of 0.746 (Kendall’s $\tau = 0.492$ [Kendall, 1938]) – comfortably clearing the 0.60 accuracy threshold and a clear improvement over both the zero-parameter tangent-rule baseline and an angle-only logistic baseline (both 0.544 accuracy, $\tau = 0.088$). The permutation test, however, gives $p = 0.27$: the significance half of the criterion fails by nearly the same margin as it does for exact-value regression. For porosity(α) ($n = 92$ rows, 20 materials, 2,408 directed pairs), the best model (a linear ranker) reaches 0.725 accuracy ($\tau = 0.450$, versus a 0.602/ $\tau = 0.204$ baseline) but again fails significance, $p = 0.28$.

This rules out “wrong task formulation” as the explanation for the pooled negative result: reformulating from exact-value prediction to relative ranking recovers strong, real-looking accuracy gains over the physics baseline in both observables, yet the same permutation test that rejects the regression models also rejects the ranking models, by almost the

Table 7: Pairwise-ranking reformulation results (pre-registered bar: pooled LOMO-CV pairwise accuracy ≥ 0.60 and permutation $p < 0.05$; $n_{\text{perm}} = 100$).

Observable	Rows	Materials	Best model	Accuracy	Perm. p	Bar cleared?
$\beta(\alpha)$	213	40	gradient-boosted ranker	0.746	0.27	No
Porosity $\phi(\alpha)$	92	20	linear ranker	0.725	0.28	No

same margin. The bottleneck is not what the model is asked to predict; it is the number of independent material-level samples available to test whether the predicted structure is more than a fold-specific artifact.

3.7 The Bayesian hierarchical partial-pooling model: a genuine complementary positive finding

As a final, deliberately different line of inquiry, the pre-registered Bayesian hierarchical model (Section 2.8) was fit to the $\beta(\alpha)$ corpus ($n = 213$ rows, 40 materials). All MCMC diagnostics are clean: maximum $\hat{R} = 1.0000$, minimum bulk effective sample size = 1342, across all seven population-level parameters.

Of the six descriptor coefficients (three each for the intercept and slope layers), one – the effect of atomic mass on a material’s mean offset from the tangent rule, mean = -3.639 , 94% HDI = $[-7.240, -0.344]$ – excludes zero. The other five (melting point and bulk density on both intercept and slope, and mass on slope) have HDIs that include zero, though several are one-sided enough to be suggestive rather than flatly null (Table 8).

Table 8: Bayesian hierarchical model: population-level descriptor coefficients (standardized descriptors), 94% highest-density intervals (HDI). Only atomic mass’s intercept-layer coefficient excludes zero.

Layer	Descriptor	Mean	94% HDI	Excludes 0?
Intercept (a_i)	Atomic mass	-3.639	$[-7.240, -0.344]$	Yes
Intercept (a_i)	Melting point	-3.288	$[-6.588, 0.151]$	No
Intercept (a_i)	Bulk density	1.121	$[-2.424, 4.884]$	No
Slope (b_i)	Atomic mass	-1.554	$[-4.916, 1.834]$	No
Slope (b_i)	Melting point	-2.552	$[-5.943, 0.475]$	No
Slope (b_i)	Bulk density	2.214	$[-1.070, 5.528]$	No

The primary deliverable, the Bayesian R^2 , is more informative than the per-coefficient

test: the posteriors are not concentrated near zero. The intercept-layer R_a^2 has posterior median 0.28 (94% HDI [0.045, 0.512]; $P(R_a^2 > 0.1) = 92\%$, $P(R_a^2 > 0.2) = 72\%$); the slope-layer R_b^2 has median 0.20 (94% HDI [0.005, 0.430]; $P(R_b^2 > 0.1) = 80\%$, $P(R_b^2 > 0.2) = 50\%$). Read plainly: it is quite likely (80–92% posterior probability) that these three descriptors explain a non-trivial share of material-to-material variation in $\beta(\alpha)$, with better-than-even odds the share exceeds a fifth – but the credible intervals are wide and do not exclude a near-zero effect at one end, and substantial residual per-material idiosyncrasy remains unexplained either way (residual scales $\sigma_a = 8.41^\circ$, $\sigma_b = 7.03^\circ$, both large relative to the mean coefficients).

This does not contradict the negative frequentist verdict of Sections 3.1–3.6, and should not be read as doing so – it answers a different question. A real-but-modest effect, imprecisely estimated (median R^2 around 0.2–0.3 but a wide HDI reaching close to zero), is exactly the kind of signal a leave-one-material-out permutation test at $n = 37$ –43 materials is underpowered to detect at $p < 0.05$ even when the effect is genuine. Two caveats apply. First, crystal structure was excluded from the group-level regression for identifiability reasons (19 levels across 43 materials); if it carries real information, as is physically plausible for a shadowing/anisotropy effect, the true descriptor-explainable variance could be higher than reported here. Second, the Bayesian R^2 reported is in-sample, not cross-validated – it measures how much structure the fitted model captures, not how well it would predict a new material’s curve.

3.8 Summary of results

Across five independent lines of attack – the pooled model at a further-grown corpus, material-class stratification, a targeted corpus-growth push confined to the most promising subgroup, a ranking/ordinal task reformulation, and a Bayesian hierarchical framework – no frequentist formulation clears the pre-registered significance bar. RMSE (or pairwise-ranking accuracy) improvement over the physics baseline is real and large in every configuration tested, from +23% to +54% depending on subgroup, and 0.72–0.75 pairwise accuracy against a 0.54–0.60 baseline for the ranking reformulation, so the learned

models consistently find genuine, reproducible structure in the data rather than overfitting to noise that happens to look good in-sample. The barrier throughout is specifically statistical power, not model quality or task framing: every reformulation tried reproduces the same pattern of strong effect size and weak significance, and the one framework built to be sensitive to a real-but-modest effect under small samples (the Bayesian hierarchical model) finds exactly that.

4 Discussion

4.1 Why this is a genuine finding, not a failed project

A reader could reasonably ask whether a paper reporting “no configuration clears a pre-registered significance bar” is worth publishing. We argue it is, for three reasons specific to this investigation’s design, not as a general defense of negative results.

First, the diagnostic completeness is itself informative. Four distinct, plausible explanations for the pooled model’s shortfall were tested directly and independently ruled out: wrong feature set (the feature-engineering ablation of Section 3.2 changed neither RMSE improvement nor p), wrong extraction method (the deep figure-re-digitization pass of the same section improved p from 0.13 to 0.09 but did not close the gap), class heterogeneity (Section 3.3, no subgroup clears the bar despite each being chemically more homogeneous than the pooled set), and wrong task formulation (Section 3.6, the ranking reformulation reproduces the identical strong-effect/weak-significance pattern). Ruling out four specific, testable alternative explanations and converging on a fifth (sample size) that is directly measurable and directly actionable is a stronger scientific contribution than an unexplained single negative result would be.

Second, the improving trend is consistent with a real, recoverable effect rather than pure noise. $\beta(\alpha)$ ’s permutation p fell monotonically with every genuine data-density increase in the corpus-growth campaign of Section 3.2 ($0.13 \rightarrow 0.09$), and the material-class-stratified subgroup closest to significance (elemental metals) is also the chemically most homogeneous one – directionally consistent with “more physically similar materials,

and more data on the same materials, moves p the right way,” even though neither trend alone reaches significance at current sample sizes. We present this as a plausibility argument for future corpus growth, not as proof the effect is real; the honest frequentist verdict at every snapshot tested here remains NOT CLEARED.

Third, and most directly, the Bayesian hierarchical analysis (Section 3.7) demonstrates *why* a permutation-test failure at $n = 14$ –43 materials does not mean there is no effect. The same corpus that fails the material-level permutation test at $p = 0.20$ –0.36 yields Bayesian R^2 posteriors with 80–92% probability of explaining more than 10% of between-material variance in $\beta(\alpha)$, concentrated specifically on atomic mass. These two results are not in tension: a permutation test at this sample size is a coarse, conservative instrument, and a real-but-modest effect of exactly the size the Bayesian model estimates is precisely what it is underpowered to detect. Reporting only the frequentist NOT-CLEARED verdict, without the Bayesian analysis that characterizes *how* partial the evidence is, would understate what this corpus actually supports; reporting only the Bayesian result, without the frequentist verdict, would overstate it. Presenting both together is, we argue, the honest way to report this class of result.

4.2 Sample-size, not extraction quality, is the binding constraint

The single most consistent pattern across every section of Section 3 is that RMSE improvement (or pairwise-ranking accuracy) is never the limiting factor – it clears its threshold in every configuration tested except $E_g(\alpha)$, which independently fails for a different, likely corpus-specific reason (only 6–10 materials, the smallest of the four observables). The limiting factor is always the number of *independent materials* available to the permutation test, which sets the resolution of its null distribution. This has a direct, testable implication: the levers that do not work (more rows on existing materials from scattered sources, different task formulations, different material-class partitions) all leave the material count roughly unchanged or reduce it; the one lever that measurably improved p within this investigation (same-source figure-level re-digitization, which both added rows and, when it introduced genuinely new materials like the Besnard densification, did so

cleanly) is the one that increases the corpus’s material diversity without simultaneously increasing between-source measurement variance. This is consistent with the cross-source-vs-same-source contrast observed directly in Section 3.4: adding W, Mo, and Au points from scattered, independent labs increased row count but *worsened* p ($0.20 \rightarrow 0.30$), plausibly because different labs’ deposition conditions (rate, substrate, base pressure) introduce real between-source variance that a single-descriptor model cannot separate from genuine material effect, whereas the single-campaign Besnard extension added new materials with internally consistent measurement conditions and moved p in the right direction relative to the cross-source attempt (though still short of the 14-material baseline).

4.3 What would resolve the open question

The material-class-stratification result identifies a promising, concrete target for future work: elemental metals reach $p = 0.20$ at $n = 14$ materials, the closest *class-stratified* configuration tested to the 0.05 bar (the pooled, all-classes $\beta(\alpha)$ fit of Section 3.1 has since moved closer still, to $p = 0.10$ at $n = 46$ materials, but that is a different, larger-sample comparison, not a same-subgroup one). Elemental metals are also the chemically most homogeneous and best-characterized class (complete, high-confidence descriptor coverage for all four descriptors used). Recovering significance in this subgroup plausibly requires roughly double the current material count – on the order of 30+ elemental metals with genuine angle-resolved $\beta(\alpha)$ data. A targeted search for 19 further elemental metals, concentrated on platinum-group and rare-earth species not yet in the corpus, returned usable data for only one (Sb) via the open web search tools available to this project; this is a practical ceiling of accessible open literature search, not evidence that the underlying data does not exist in the broader, harder-to-reach literature (older or non-open-access journals, conference proceedings, and materials-science theses not indexed by general web search). Access to a professional literature database with more complete coverage of these specific material families (Scopus, Web of Science, or equivalent) is therefore the single most direct, well-motivated path to resolving whether the elemental-metals signal is real. We flag this explicitly as the concrete deliverable this paper offers to the next phase of this

research program, rather than leaving “more data would help” as an unspecific closing remark.

4.4 Limitations

Several limitations apply beyond the sample-size discussion above. The corpus is drawn from heterogeneous deposition conditions (rate, base pressure, substrate, chamber geometry) that are not uniformly reported across source papers and are only partially captured by the deposition-method categorical descriptor; residual between-source variance not attributable to material identity is folded into what the permutation test treats as noise, which could mask a real effect or, less plausibly, contribute to the apparent RMSE improvements through incidental correlation with material identity. The Bayesian hierarchical model’s R^2 is in-sample, not predictive, and excludes crystal structure for identifiability reasons; both caveats mean its positive finding should be read as a characterization of the corpus’s explainable structure, not a validated predictive claim. Finally, all results here concern angle-resolved morphological observables (β , porosity) under a single physics baseline (the tangent rule); the two companion observables tested (refractive index, band gap) are optical rather than morphological and were evaluated primarily to check whether the negative pattern was specific to β /porosity or general across GLAD-relevant observables – it proved general, which we read as additional, independent evidence that the underlying constraint is sample size rather than an observable-specific modeling failure.

5 Conclusion

We conducted a pre-registered, leakage-controlled evaluation of whether material descriptors (atomic mass, melting point, bulk density, crystal structure, deposition method) can predict a new material’s column-tilt angle $\beta(\alpha)$ or porosity $\phi(\alpha)$ under glancing-angle deposition, using leave-one-material-out cross-validation and a material-level permutation test as the significance instrument. Every learned model comfortably clears the pre-registered RMSE-improvement bar for $\beta(\alpha)$ (up to +54.4% depending on corpus snap-

shot) and usually for porosity (up to +28.5%), but no configuration tested – the pooled fit across up to 46 materials, three pre-registered material-class subgroups, two independent targeted corpus-growth interventions, or a pairwise ranking reformulation of the task – ever clears the permutation-test significance bar. Four plausible alternative explanations (feature choice, extraction quality, class heterogeneity, task formulation) were tested directly and ruled out; a fifth – that the negative results are an artifact of per-test underpowering rather than a genuine absence of effect – was tested via a pre-registered Fisher’s combined-probability test across the three disjoint material-class subgroups and separately across the three pooled observable-level tests, and also ruled out (combined $p = 0.715$ and $p = 0.121$ respectively, the latter updated 2026-08-21 after the second same-day corpus-growth round dropped its two beta/porosity inputs to $p = 0.10/0.12$; neither combination clears the bar, though the observable-level combination’s input p-values now sit closer to the diffuse-weak-signal regime Fisher’s method is designed to detect than at any earlier snapshot). The specific, remaining, and well-quantified bottleneck is material-level sample size. A complementary Bayesian hierarchical partial-pooling model, fit independently and asking a different statistical question, finds a genuine, credible non-zero effect of atomic mass on the material-level offset from the physics baseline and places 80–92% posterior probability on descriptors explaining more than 10% of between-material variance – a result fully consistent with, and explanatory of, the frequentist non-significance verdict rather than contradicting it.

We report this as a rigorous, multi-angle-tested negative-to-mixed result: a well-instrumented account of exactly where the statistical-power boundary for cross-material GLAD morphology prediction currently sits, together with the specific, actionable corpus-growth target (professional literature-database access to 30+ elemental metals, particularly platinum-group and rare-earth species) most likely to move it. We offer this both as a direct contribution – ruling out four alternative explanations for a persistent gap between effect size and statistical significance in small-corpus materials-informatics problems is itself useful to readers attempting similar cross-material generalization studies – and as a resource for future corpus-growth efforts specifically in this sub-field, so that the

620 next research effort in this direction can begin from a documented, tested starting point
621 rather than repeating the same exploratory search.

622 **Data and Code Availability**

623 All feature tables, model-comparison scripts, and pre-registration documents referenced
624 in this paper are maintained in the project repository (02_PINN_NIF/P3_ML/); the corpus
625 is drawn from peer-reviewed literature and PhD theses, cited where individual source
626 papers contributed the majority of a material’s data (Section 1), with the full per-row
627 provenance recorded in the underlying feature-table generation pipeline.

628 **CRedit Author Statement**

629 Oussama Ben Haj Ali: conceptualization, methodology, software, formal analysis, investi-
630 gation, data curation, writing – original draft, writing – review and editing, visualization.
631 Ferid Chaffar Akkari, Bruno Gallas, Nicolas Martin: resources, supervision, writing –
632 review and editing.

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